

Gel Filtration Procedures

2024

Proteins and other macromolecules can be separated based on molecular weight (MW) by using a cross-linked porous gel. Gels have various degrees of cross-linking that allow certain sized molecules to pass through while others cannot. The degree of retardation of a particle is related to the molecular weight and shape. Some molecules will be excluded from the gel, and others will be included. Those that are excluded will pass through the column faster because the distance they have to travel is reduced. The smaller molecular will take a more convoluted path through the column and elute later.

1. Buffer preparation:

Buffer A: 50 mM Phosphate Buffered Saline (0.14 M NaCl, 36 mM Na₂HPO₄, 14 mM NaH₂PO₄), pH 7.2. Filter buffer A using 0.45 µm filter before use.

In Gel Filtration experiment, only buffer A will be used. Clean buffer bottle A and add 400 mL Buffer A to bottle A. Immerse buffer A inlet tubing and buffer B inlet tubing in bottle A.

2. Switch on AKTA start and install pump tubing. Connect computer to AKTA start.

(Step 3 and 4 may be finished by TAs or volunteers.)

3. Start manual run through system control: wash buffer A tubing (20 mL); wash buffer B tubing for (20 mL); wash fractionation tubing (10 mL); extrude air bubble in tubing. Flow rate: 5 mL/min.

4. Connect Sephacryl S-100 to the instrument. Flow rate: 1 mL/min.

5. Equilibration: 0.5-1 CV. Flow rate: 1 mL/min

6. Prepare sample and filter sample using 0.45 µm filter.

Sample containing:

Blue dextran 2000	5 mg/mL	MW: 2000 kDa	(For V ₀ measurement.)
BSA(Bovine albumin)	6mg/ml	MW: 67 kDa	(Standard protein 1)
Ovalbumin (chicken)	9mg/ml	MW: 45 kDa	(Standard protein 2)
Unknown protein	8 mg/mL	MW: ?	
Ribonuclease A (Bovine)	9mg/ml	MW: 13.7 kDa	(Standard protein 3)

7. Wash inlet tubing and sample vale. Sample application: sample valve-0.2-0.5 mL, flow rate: 0.8 mL/min.

Note: Always make sure the state of each valve and content in tubing in programed activity of chromatography system. Avoid air bubble enter fractionation tubing. Pause the program or switch wash valve connection to waste if necessary; switch sample valve connection to buffer quickly to prevent air enter column after sample application in manual run.

8. Elution and collection: using buffer A elute proteins, flow rate: 0.8 mL/min. In this experiment, all eluted proteins are **not required** collection. Elution setting: make sure switch outlet valve connection to waste. When all proteins are eluted, stop the program

after additional 15 min-elution.

9. Export csv file (or save pdf file if necessary) and upload through local area network (NOWLab). Make a graph of elution curve and find V_0 and elution volume of each protein. Calculate K_{av} of each protein and plot logMW versus K_{av} form a standard curve. Estimate molecular weight of unknown protein through standard curve.

(Step 10 to 12 will be finished by student in the last class or volunteers.)

10. Column cleaning: 0.2 M NaCl, 0.5 CV; water, 1CV. Flow rate: 1 mL/min.
11. Column regeneration: 0.5 M NaCl, 1 CV; 0.1 M HOAc, 1CV (or 0.2 M NaOH). Flow rate: 1 mL/min.
12. Column cleaning: water, 1CV; 20% alcohol, 1CV for storing. Flow rate: 1 mL/min.

[Watch the video in WEB LEARNING and go through the procedures before class.]

[Experiment report for GF is **not required**. Calculation will be finished in **Assignment 1**.]